



Heterogeneous preferences for social trail use in the urban forest: A latent class model



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ARTICLE INFO

Article history:

Received 2 September 2015

Received in revised form 23 June 2016

Accepted 23 June 2016

Available online 24 June 2016

Keywords:

Discrete choice experiment

Harvesting wild food plants

Motivations

Hokkaido, Japan

Nopporo forest park

ABSTRACT

This study explored visitors' preferences for social conditions of trail use in urban forests. It is important to understand the heterogeneous nature of visitor preferences because the recreational use of urban forests is characterized by multipurpose uses with different visiting motivations. To understand preference heterogeneity, this study used a discrete choice experiment using a latent class model that incorporated motivational and sociodemographic factors as membership likelihood function. In 2006, 256 visitors to the Nopporo Forest Park, Hokkaido, in northern Japan, completed questionnaires with a series of discrete choice tasks using computer-manipulated images presenting various social conditions of trail use, such as visitor numbers and harvesting behavior on wild food plants. The study identified two visitor groups. One group consisted of more elders whose motivation was to enjoy the natural environment. Although they tended to tolerate crowded conditions, they particularly disliked plant pickers, who harvest wild vegetables alongside the trail. The other group was less tolerant of crowds, and enjoying the natural environment was not their primary purpose for visiting the park. These visitors especially disliked excessive numbers of visitors, but they did not dislike plant pickers. These results suggest that traditional park management assumptions about typical visitors poorly serve the heterogeneous nature of the visitor population.

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1. Introduction

The few urban natural areas that still remain in densely populated regions are **valuable because they provide many opportunities for city-dwellers to escape from hectic urban life** (Hammitt, 2002). At the same time, these areas are heavily used by a large number and a wide variety of visitors. **Crowding and visitor conflicts** can arise and may affect the quality of recreational experiences. The former is defined as a negative evaluation of a certain number of people in a given situation, and the latter as goal interference attributed to other users' behavior or social value differences (Manning, 2010). Thus, one of the key challenges in urban forest management is to provide appropriate recreational opportunities with the **most desirable trail conditions for visitors** (Dwyer et al., 1992; Arnberger et al., 2010). To address this issue, information on visitors' preferences for social conditions of trail use (e.g., use lev-

els, types of use, visitor behavior) based on their visiting motives is needed. Motivation has been found to be important in explaining differences in visitors' preferences and behaviors (Manning, 2010).

Urban forests serve multiple purposes for their visitors (Dwyer et al., 1992). For example, Hammitt (2002) revealed that visitors **seek solitude; however, as demonstrated by Vaske and Shelby (2008), levels of use affect solitary visitors to varying degrees according to the country, region, and type of activity**. People living in Asian countries, where our study area is located, generally tolerate congestion, due to their crowded living conditions according to Gillis et al. (1986) and Homma (1990). However, it should not be assumed that Asians are uniformly tolerant of congestion: Arnberger et al. (2010) categorized visitors to urban forests in both Sapporo, Japan and Vienna, Austria into subgroups according to preference for congestion. This indicates that visitors are probably heterogeneous in their preferences for social conditions in urban forests, where various activities occur for different purposes.

This study uses an image-based discrete choice experiment (DCE, hereafter) to examine visitors' preferences, and any heterogeneity, for social conditions in trail use in the Nopporo Forest Park

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(NFP), Hokkaido, in northern Japan. Little is currently known in Japan about how visitors' motivations and sociodemographic characteristics relate to such preferences, and so this study aims to: (1) understand heterogeneity in preferences for trail use in urban forests, (2) identify the factors for preference heterogeneity, and (3) derive practical management recommendations.

DCE was initially developed by Louviere and Hensher (1982) and Louviere and Woodworth (1983), and is one option from a family of stated preference approaches (Louviere et al., 2000). In addition to recreation management, the method is widely used in marketing and transportation. In our case each respondent received four choice sets, each containing three alternative trail scenarios with attributes (i.e., social conditions of trail use), from which they selected their preference; the random utility approach was then used to formulate respondents' choice behaviors. Once one understands how changes in attributes influence behaviors and/or satisfaction levels through DCE, one can predict how they will be affected by different levels of attributes (Louviere and Timmermans, 1990).

To achieve the above-mentioned research aims, DCE was used with a latent class model (LCM, hereafter). **LCM not only elicits heterogeneous preferences but also identifies the sources of heterogeneity in a single estimation framework. Identifying the sources of heterogeneity can be useful to land managers in predicting what trail conditions or uses are preferred by specific groups, through observing their characteristics.** Thus, LCM makes an important contribution to the literature of urban forestry management. As a point of comparison, a conditional logit model (CLM, hereafter) estimation was also undertaken.

Our analysis is also unique in **identifying visitors' preferences for harvesting behaviors of edible plants in urban forests.** To the best of our knowledge, there was no study that tried to understand preferences of harvesting (or poaching) behavior and their trade-offs among other social variables such as visitor numbers and visitor behaviors. Harvesting wild food plants such as berries, mushrooms and mountain vegetables are popular and usual recreation activities in the world (e.g., Bell et al., 2007). In Japan, people have traditionally harvested wild plants and mushrooms. While these plants were important parts of the rural diet in the past, today, **the harvesting of edible plants and mushrooms by city dwellers is almost exclusively considered a recreational activity (Saito, 2001).** On the other hand, **harvesting is also a major challenge in some urban forests, since natural resources may be exhausted by consumptive recreational harvesting by city dwellers.** Moreover, even when harvesting is not so severe, such behavior may produce conflict with visitors who have other purposes.

1.1. Review on visitor preferences

Over the past few decades, a large number of studies have been conducted on recreational preferences, especially in backcountry areas. **The effect of crowding on visitors' recreational experiences is one of the most intensively studied issues, as crowding is considered to have the most direct and significant social impact (e.g., Graefe et al., 1984; Shelby and Heberlein, 1986; Manning et al., 1999; Vaske and Shelby, 2008).** The earliest hypothesis claimed that the perception of crowding is predominantly affected by the number of encounters one has with other recreationists. However, later studies have found that crowding is an individual, subjective judgment and that variability across individuals is attributable to social, psychological and situational factors (Absher and Lee, 1981; Ditton et al., 1983; Shelby and Heberlein, 1986). That is, in addition to use level in a recreational setting, other factors such as the type of use and other visitors' behaviors need to be considered when we examine visitors' preferences for social conditions of trail use. For example, previous studies show that unwanted behaviors

such as littering, letting dogs off their leashes and the size of other visitor groups also influence outdoor recreation quality (Altman, 1975; Manning and Valliere, 2001; Heywood and Murdock, 2002; Arnberger and Haider, 2005).

Nonetheless, to date, only a few studies have addressed preferences for social conditions of trail use in urban forests (Arnberger and Haider, 2005; Arnberger et al., 2010; Arnberger and Eder, 2011). Moreover, past studies have focused on urban forests in Central European countries and the USA. To our knowledge, there have been few such studies in Asian countries, although a notable exception is the study of Koo et al. (2013). Findings in European countries and USA may not generalize to Asian recreational settings, as previous studies have found that Asian people have different preferences in recreational settings. (e.g. Sayan et al., 2013; Arnberger et al., 2010). Therefore, preferences for social conditions of trail use and specific visitor behaviors in urban forests remain rather unknown in Asian countries.

1.2. Analysing preference heterogeneity

While DCE is powerful in eliciting the impact of social attributes on visitors' satisfaction, it is not straightforward in modeling preference heterogeneity. For example, some studies first conduct a multivariate cluster analysis of individual-specific characteristics (e.g., psychometric or socioeconomic), and then estimate choice behavior by cluster. However, unlike LCM, clusters are based only on characteristics and not preferences; consequently, there may be multiple clusters with different visitor characteristics but similar preferences. LCM, though, avoids this by basing clusters on both characteristics and preferences, which helps the identification of characteristics responsible for preference heterogeneity.

Difficulty in understanding preference heterogeneity is due to modeling individual-specific characteristics, which are considered to be the main source, but are canceled out in the context of DCEs within a random utility framework. Previous studies therefore introduced interaction terms between individual-specific characteristics and choice attributes under the CLM framework (e.g., Morey et al., 2002; Arnberger and Haider, 2005); however, this is limited because it requires a priori selection of key individual characteristics and attributes. Another way of modeling preference heterogeneity is the random parameter logit models (e.g., Train, 1998), but it is not well suited to explaining the sources of heterogeneity. In contrast, LCM not only elicits preference heterogeneity but also identifies its sources in a coherent framework. Over the past few decades, especially in the field of marketing, studies have focused on modeling preference heterogeneity or identifying consumer segmentation for goods and services (e.g., Wedel and Kamakura, 2000), and this study follows this methodological trend.

In the literature on recreation, LCM has been applied to model choice behaviors in wilderness parks and wildlife tourism (Boxall and Adamowicz, 2002; Semeniuk et al., 2009), while Arnberger et al. (2010) and Reichhart and Arnberger (2010) used LCM to understand preferences for social conditions of trail use in urban forests. Arnberger et al. (2010) first estimated an LCM without modeling how visitors' characteristics affected preference heterogeneity, and then made post-estimation comparisons of visitors' characteristics across groups to identify the sources. This study takes a different approach by explicitly modeling group membership as a function of visitors' motivations and sociodemographics.

2. Method

2.1. Study area

The study area was the NFP, which is east of the City of Sapporo. With a population of about 1.9 million people, Sapporo is the fifth

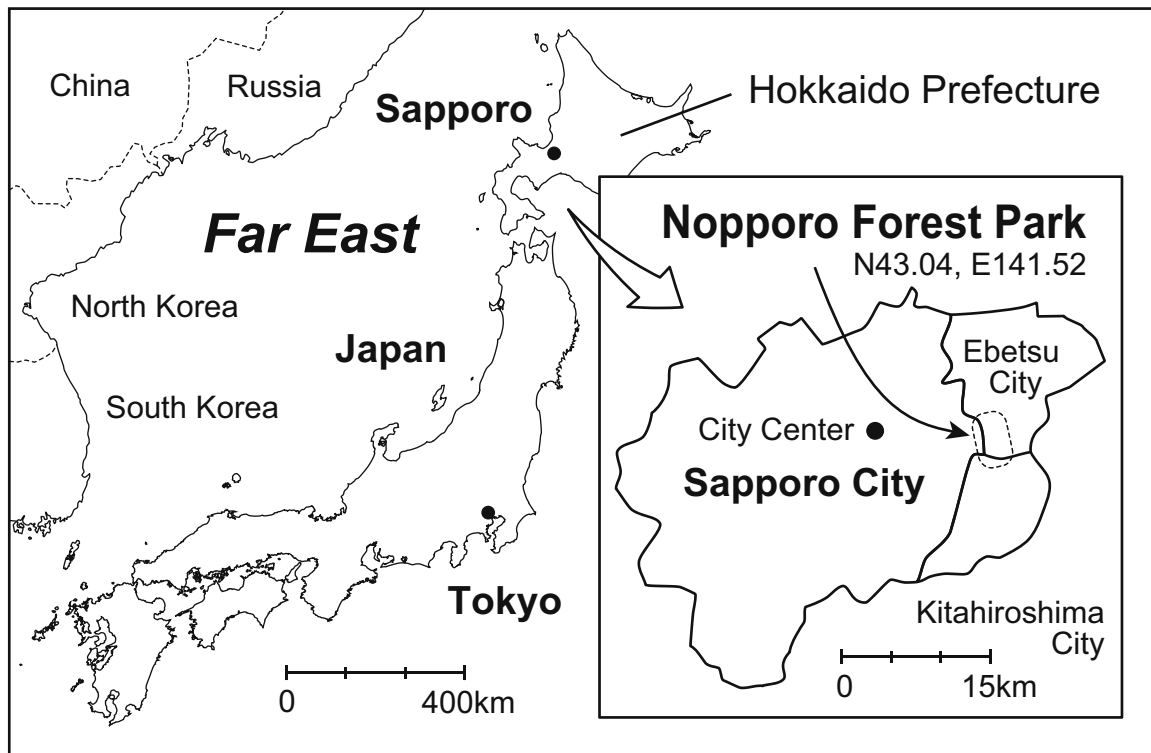


Fig. 1. The location of Nopporo Forest Park, Hokkaido, Japan.

Table 1
Results of the factor analysis of visitor motivations.

Motivations for urban forest use	Mean ^a (s.d.)	Factor 1	Factor 2	Factor 3
Appreciation of the natural environment				
Relaxing or recreation	3.308 (0.782)	0.588	0.115	0.218
Appreciating the scenery	3.288 (0.760)	0.699	0.075	0.181
Understanding nature in the forest	2.446 (1.083)	0.673	0.149	−0.065
Appreciating plants and animals of the forest	3.403 (0.753)	0.682	0.165	0.105
Feeling solitude				
Longing for solitude	2.322 (1.133)	0.166	0.735	0.164
Practicing meditation	1.883 (1.057)	0.177	0.723	0.178
Exercise or participates in sports				
Exercising for health	3.464 (0.745)	0.229	0.078	0.647
Participating in sports	3.099 (1.017)	0.007	0.267	0.599
Eigenvalues		3.015	1.430	1.034

^a Likert scale (from unimportant = 1 to important = 4).

largest city in Japan and the prefectural capital of Hokkaido (Fig. 1). The forest, which was designated as a protected area (Hokkaido prefectural natural park) in 1968, covers an area of 2051 ha and is the largest of the parks located on the outskirts of big cities in Japan.

The forest occupies an important position among the recreational areas in this city suburb, and it has several facilities such as a visitor center, a museum, about 40 km of maintained hiking trails and several parking lots. The park had approximately 659,000 visitors in 2005 and, in 2006, about 80% of the visitors were walkers, followed by bicyclists, joggers and dog walkers (Arnberger et al., 2010).

The main management issues in the NFP are overuse and off-trail use including harvesting wild vegetables. Some trails leading out of the Osawa trailhead, which is a main access point of the NFP, provide a good natural environment for nature walking and bird watching, as well as dog-walking and jogging opportunities for residents from the nearby area. A typical off-trail activity is picking wild vegetables. Even though picking vegetables is not illegal in the NFP (except for some designated species), because the activity is

based on vested rights regarding traditional forest uses, area managers always appeal explicitly to visitors to refrain from harvesting wild plants in the NFP.

2.2. Data sampling

On-site interviews were conducted at the two main access points: Osawa and Kinentoguchi trailheads on **two randomly selected Fridays and two randomly selected Saturdays during June and July of 2006**. We chose this period because it is ideal for surveying harvesting behavior. This short sampling period allowed us to draw conclusions just for visitors in the early summer period.

Most of the interviewers were students, and they were thoroughly instructed in the use of the survey forms. They asked visitors if they would be willing to participate in a **15-min interview**. **During the survey period, 325 of the 401 (81.0%)** visitors solicited agreed to participate. After excluding surveys from respondents who did not complete the questions needed for LCM, 256 completed surveys remained.

Table 2
Attributes and levels in the discrete choice experiment.

Attributes	Levels
Number of walkers	2 walkers 5 walkers 10 walkers 15 walkers
Number of bikers	No biker 1 biker 2 bikers 3 bikers
Number of joggers	No jogger 1 jogger 2 joggers 3 joggers
Plant picker	No plant picker 1 plant picker
Dog walker	No dog walker 1 dog walker with leashed dog 1 dog walker with unleashed dog

The interviewers collected visitors' sociodemographic characteristics and asked visit-related questions regarding the length and frequency of their visits, as well as questions about their motivations for using the urban forest. Eight reasons such as "appreciating plants and animals," "exercising for health," or "longing for solitude" were recorded using a Likert scale (Table 1). These eight variables were factor-analyzed by principal component analysis with Varimax rotation (we will deal with the results in a latter section).

Then interviewers asked respondents to evaluate **twelve trail use scenarios**. These scenarios consist of images created using the Photoshop software, which were organized into four choice sets. **Image-based techniques have been one of the primary methods for eliciting visitors' recreational preferences** (e.g., Hof et al., 1994; Manning et al., 1996). Several studies have advanced the image-based technique by using a DCE to understand trade-offs among different social attributes such as use level and user conflicts such as littering and unleashed dogs (e.g., Lawson and Manning, 2002; Arnberger and Haider, 2005; Arnberger et al., 2010). Trail preferences were assessed by asking visitors to choose their most, second most and least preferred scenario of the three scenarios depicted in each set. Although we could have used a rank-order interpretation of our data, we considered it as a choice task and used the first-choice information rather than all three alternatives.

2.3. Trail scenario design

We created the trail scenarios with Adobe Photoshop. The background of the images was a section of a trail near the Osawa trailhead. Alternative trail scenarios were created by manipulating five attributes and their levels (attribute ranges) of social conditions of trail use (Table 2 and Fig. 2).

- (1) Total number of walkers
- (2) Number of bikers
- (3) Number of joggers
- (4) Presence/absence of a plant picker
- (5) Presence of dogs and whether or not they are leashed

Attribute levels were set according to interviews with area managers, a pretest conducted in May of 2006 and the currently observed pattern of trail use in the NFP (Kaise et al., 2007). Walking is a major form of recreation in the NFP; thus, it is natural for respondents that images of walkers constitute a large proportion of figures in trail scenarios. To this end, the levels of the attribute were

set higher than those of bikers and joggers. We set the maximum number of people at 15 in order to examine how people respond to this now-rare but future-possible situation.

Although bikers and joggers do not currently produce serious conflict between visitors in the NFP, these attributes were included as a possible cause of future conflict. It is very rare to see more than 3 bikers or joggers on that trail. Therefore, 3 was chosen as the maximum number of visitors for those activities, respectively. For plant picker, we set its maximum attribute level at 1 as one rarely see plant picking in the park. The attributes presence of a plant picker and a dog walker are included in the model as dummy variables with effects coding (Louviere et al., 2000). Dog walking can be seen occasionally and thus we set its attribute level to whether it is present or not. A dog can be leashed or unleashed when a dog walker is present. Walking with unleashed dogs in large cities is prohibited in Japan, including Sapporo. Nonetheless, a few dog walkers still unleash their dogs.

In this study we first created 216 scenarios based on the full factorial design. Then, we removed images that could not be technically processed by researchers. Consider a scenario where the maximum number of figures from each attribute is placed in a picture. The image is so congested with people on top of each other on the 2-dimensional space that it is impossible for respondents to recognize the attribute of each person in the image. In order to avoid this phenomenon, we took out combinations of attributes that could pose difficulty in identifying the attributes and their levels to the respondents. This reduced the number of designs to 78 for which we are confident that each person's attribute is clearly identifiable. This procedure left us a non-orthogonal design and generally leads to loss in statistical efficiency. Moreover, dropping some scenarios is commonly done to enhance the quality of response and thus the response efficiency (e.g., Johnson et al., 2013).

2.4. Description

Both CLM and LCM build on the well-known random utility framework. It assumes that the utility function is the sum of a deterministic term that can be described as a function of factors that influence respondent n 's utility and a random term that is unobservable and stochastic. Utility for a profile i is described in Eq. (1):

$$U_{ni} = V_{ni} + \varepsilon_{ni}, \quad (1)$$

where U_{ni} is the total utility for profile i , V_{ni} is an observable deterministic term and ε_{ni} is an unobservable random term.

A CLM (McFadden, 1973) assumes that the error term is distributed as Type I extreme values. Then, the probability that profile i is chosen can be expressed as Eq. (2):

$$P_n(i|C) = \exp(\mu\beta x_i) / \sum_{j \in C} \exp(\mu\beta x_j). \quad (2)$$

We model the observable part as $V_{ni} = \mu\beta x_i$, where x_i is a vector of attributes of choice i that affect individual n 's utility, β is the coefficient parameter of attributes, and μ is the scaling parameter. In this study, x_i consists of attributes that are detailed in Table 2. Generally, the scale parameter μ is assumed equal to 1. While the basic structure of the LCM is similar to the CLM, it adds more flexibility by allowing for heterogeneous preferences by groups that are a priori latent. Our LCM was originally proposed by McFadden (1986) and first implemented by Swait (1994). Following Swait (1994), Boxall and Adamowicz (2002) and Ruto et al. (2008), consider a latent membership likelihood function $M^*_{n|s} = \gamma_s z_n + \zeta_{n|s}$, where z_n is the motivational as well as sociodemographic characteristic of individual n , γ_s is a vector of parameters and $\zeta_{n|s}$ is an unobservable random term. The error terms are assumed to be independently distributed across individuals and segments with Type I extreme value distri-

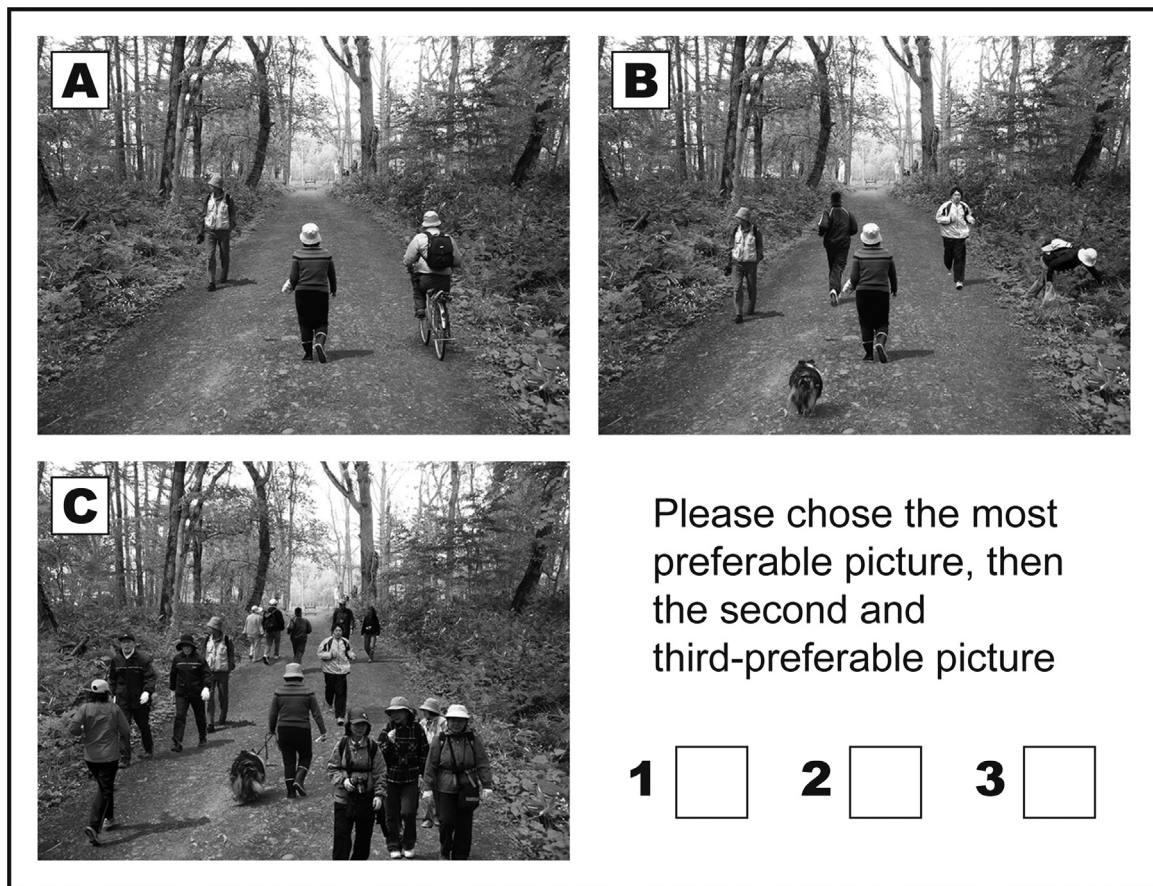


Fig. 2. An example of a choice set of the discrete choice experiment.

bution, and the probability of a sequence of choices by respondent n in segment s can be expressed as follows:

$$P_n(i_1, i_2, i_3, i_4|s) = \sum_s P_{n|s} \times \prod_{t=1}^{t=4} P_{n|s}(i_t|s) \quad (3)$$

$$= \sum_s \frac{\exp(\lambda_s \gamma_s z_n)}{\sum_s \exp(\lambda_s \gamma_s z_n)} \times \prod_{t=1}^{t=4} \frac{\exp(\mu_s \beta_s x_{i,t})}{\sum_{j \in C} \exp(\mu_s \beta_s x_{j,t})}$$

where λ_s is the scale parameter of segment s . The Wald test was used to test the significance of the sets of parameters, and whether coefficients are equal among classes (Vermunt and Magidson, 2005).

Fig. 3 shows our research framework based on Swait (1994) and Boxall and Adamowicz (2002). The white and gray boxes demonstrate observable and unobservable variables, respectively, and the flow enclosed by a heavy line indicates the above-mentioned estimation. The latent membership likelihood function contains two motivational dimensions and two sociodemographic variables. The term 'membership likelihood function' as defined here (e.g., Swait, 1994; Boxall and Adamowicz, 2002) is also called 'covariates', which are used to predict class membership (Vermunt and Magidson, 2005).

3. Results

3.1. Visitor characteristics

Although the male–female ratio of respondents was nearly balanced (males and females accounted for 56.1% and 43.5%, respectively), respondents 50 years and older constituted almost

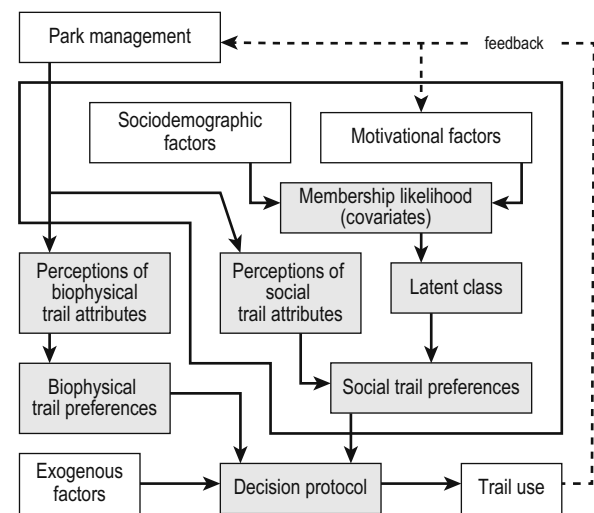


Fig. 3. Framework of our study and model of latent class.

80% of respondents (those 50–59, 60–69 and over 70 years old accounted for 24.3%, 34.9% and 19.9%, respectively). The overwhelming proportion of elderly visitors is not due to sampling bias. Aging has been advancing rapidly in Japan, and elderly populations who live nearby urban forest parks are the majority of the park visitors in Japan, as shown in Aikoh et al. (2006) as well. For about three-fourths of the respondents, it takes less than 30 min to the NFP by foot from their homes. In addition, almost two-thirds of the respondents visited the NFP more than or equal to 10 times in the previous year, of which 31 respondents (10.3%) did more

Table 3
Conditional logit results.

Variable	Conditional logit model			
	Linear model ^c		quadratic model	
	Coeff.	Wald stat.	Coeff.	Wald stat.
Number of people	−0.199	189.79**		
Quadratic term of number of people ($\times 10^{-1}$)			−0.101	211.24**
Number of bikers	−0.126	9.22**	−0.363	22.49**
Quadratic term of number of bikers ($\times 10^{-1}$)			0.540	12.04**
Number of joggers	−0.053	2.37	−0.066	4.29*
No plant picker ^a	0.229	16.95**	0.265	26.47**
One plant picker ^a	−0.229		−0.265	
No dog walker ^a	0.628	83.17**	0.595	90.22**
Dog walker with 1 leashed dog ^a	−0.280		−0.281	
Dog walker with 1 unleashed dog ^a	−0.347		−0.314	
Number of respondents	256		256	
Observations	1022		1022	
Log-likelihood (max)	−766.188		−764.418	
ρ^2 ^b	0.347		0.350	

^a Effect-coded variables.^b ρ^2 : $1 - \text{Log-likelihood (max)}/\text{Log-likelihood (0)}$.^c A goodness-of-fit measure ρ^2 of the rank-ordered logit model is 0.145.* $p < 0.05$.** $p < 0.01$.

than or equal to 200 times. Some respondents spent time at the NFP for less than one hour; other respondents did for more than two hours. The results showed that weekday visitors were more the frequent visitors (t -test, $p < 0.001$) with higher proportions of homemakers (37.7% of weekday visitors compared to 11.4% of holiday visitors). However, no statistical differences were found for other visitor characteristics (e.g., age, gender and time spent in the NFP).

3.2. Motivational dimensions

Respondents scored highest on the motive of “exercising for health,” and lowest on “practicing meditation” (Table 1). Yet, “appreciating plants and animals of the forest,” “relaxing or recreation,” and “appreciating the scenery” were also strong motives. Factor analysis identified three components, explaining 68.5% of the variance. The first factor was defined as “appreciation of the natural environment,” because “appreciating the scenery” and “appreciating plants and animals of the forest” loaded highly on this factor. The second and third factors were defined as “feeling solitude” and “exercises or participation in sports,” respectively.

3.3. Conditional logit result

Table 3 shows the estimation results of the CLM. In our analysis, we included the attribute “total number of people,” which is defined as the total number of figures shown in a computer-manipulated image, instead of the number of walkers, because we were interested in the impact of the total number of people regardless of the type of visitors. In most cases, the number of walkers accounted for most figures in an image; thus, the attribute of the number of walkers was excluded from our analysis, avoiding multicollinearity. The linear model includes only linear terms, and the quadratic model includes linear and quadratic terms. However, the linear term “total number of people” was excluded in the quadratic model because both linear and quadratic coefficient parameters were insignificant in a model with both terms. Considering the goodness-of-fit measures, we only examined the quadratic model in the remainder of this paper. The alternative-specific constants were excluded from the model because the DCE contained unlabeled alternatives (Hensher et al., 2005). We assume that the alternative without people (only background image) is the common anchoring point.

Table 4

The number of segments and information criteria.

The number of segments	AIC ^a	AIC3 ^b	crAIC ^c	BIC ^d
1	1542.8	1549.8	1574.7	1567.7
2	1470.8	1489.8	1557.1	1538.1
3	1467.9	1498.9	1608.8	1577.8

^a AIC: $-2LL(\text{max}) + 2k$ (k : the number of parameters).^b AIC3: $AIC3: -2LL(\text{max}) + 3k$.^c crAIC: $AIC + 2k(k+1)/(n-k-1)$ (n : the number of observations).^d BIC: $-2LL(\text{max}) + k \ln(n)$.

All attributes significantly influenced visitors' choices. The coefficient for the quadratic term of the total number of people was negative; the respondents indicated that the desirability of the condition decreased at an increasing rate as the numbers of people increased. The coefficients for the linear and quadratic term, number of bikers, were significantly negative and positive, respectively. The first biker decreased respondents' utility by $-0.363 \times 1 + 0.054 \times 1^2 = -0.309$, but the second and third biker did additionally so by $(-0.363 \times 2 + 0.054 \times 2^2) - (-0.309) = -0.201$ and $(-0.363 \times 3 + 0.054 \times 3^2) - (-0.201) = -0.093$, respectively. That is, the desirability of the condition decreased as the numbers of bikers increased. Thus, the negative impact of the number of bikers was limited to this range. Similarly, the number of joggers seems to have a small negative impact on visitors' utility, as the coefficient was small although the Wald statistics suggested significance. The presence of a plant picker negatively affected visitors' experience, and the absence of dogs was preferred. However, the CLM assumes that the effect of each variable on an individual's experience is constant across individuals.

3.4. Latent class result

To specify the appropriate number of segments, the information criteria of the models with different solutions were compared. As can be seen in Table 4, AIC suggested three as the optimal number of segments, whereas the AIC3, crAIC and BIC supported a two-segment model. In this study, we used two as the optimal number of segments, since AIC is known to be excessively lenient to additional parameters. The results of LCM with two segments are shown in Table 5.

Table 5
Latent class result.

Variable	Latent class model (two-segments)			
	Segment 1 (share: 57.7%) Coeff.	Segment 2 (share: 42.3%) Coeff.	Wald stat.	Wald (=) stat. ^c
Quadratic term of number of people ($\times 10^{-1}$)	−0.121	−0.584	53.69**	26.07**
Number of bikers	−0.343	−0.343	13.01**	–
Quadratic term of number of bikers ($\times 10^{-1}$)	0.460	0.460	6.00	–
Number of joggers	−0.007	−0.231	6.04*	4.46*
No plant picker ^a	0.418	−0.290	32.13**	7.76**
One plant picker ^a	−0.418	0.290		
No dog walker ^a	0.753	0.753	91.70**	
Dog walker with 1 leashed dog ^a	−0.331	−0.331		
Dog walker with 1 unleashed dog ^a	−0.422	−0.422		
Membership likelihood variables				
Constant	0.076	−0.076	0.14	
Appreciation of natural environment	0.321	−0.321	7.22**	
Feeling Solitude	−0.251	0.251	4.33*	
Respondents 60 years and older	0.442	−0.442	4.90*	
Relative low frequency of visits (less than 10 times last year)	−0.443	0.443	4.68*	
Number of respondents			256	
Observations			1022	
Log-likelihood (max)			−16.380	
ρ^2 ^b			0.468	

^a Effect-coded variables.

^b ρ^2 : 1−Log-likelihood (max)/Log-likelihood (0).

^c The Wald statistics labeled ‘Wald (=) stat.’, which tested whether coefficients are equal between classes.

** $p < 0.01$.

* $p < 0.05$.

The latent class results improved the goodness-of-fit ρ^2 , which provided evidence of heterogeneous preferences. Two motivational dimension and two sociodemographic characteristics differed between the two segments. The estimated parameter of “exercises or participates in sports” was not statistically significant in the segmentation. The sum of coefficients of each membership likelihood variable is set to zero for identification.

Segment 1 consisted of 57.7% of the respondents. Their positive coefficients of “appreciation of the natural environment” and “Respondents 60 years and older” indicated that segment 1 represents elderly visitors who were interested in enjoying the natural environment, compared with segment 2. Segment 1 was not bothered much by the total number of people, since the absolute value of the coefficient for the total number of people is smaller than that for the segment 2. However, the presence of a plant picker strongly influenced segment 1 group’s experiences. This finding is consistent with the segmentation, because one of the major motivations of the segment 1 was to appreciate the natural environment; thus, observing someone picking plants conflicts with their purpose for being in the forest.

Segment 2 consisted of 42.3% of the respondents. Segment 2 group’s positive coefficients of “feeling solitude” and “relatively low frequency of visits” indicate that segment 2 represents less frequent visitors who are interested in solitude, also compared with segment 1. In contrast to the segment 1, they were negatively affected by the total number of people. This is the same trend for the mean respondent estimated by the CLM. However, the absolute value of the latent class parameter is much larger than that in the CLM, if we assume that the alternative without visitors is the common anchoring point in both models. The parameter of the number of joggers has the same tendency as the total number of people. That is, they are much more sensitive to crowding than the average visitor. This finding is also consistent with the segmentation, because one of the major motivations of the segment 2 was to experience solitude; thus, crowded conditions conflicts with their purpose for being in the forest.

On the other hand, although the absolute values of the preference parameters for dog walkers (leashed or unleashed) were not

different between the groups represented by the segments, the trend to dislike the presence of dog walkers was common.

4. Discussion

4.1. Importance of preference heterogeneity

Our finding that a larger number of people negatively affects respondents’ experiences corresponds to that reported for Central Europe (Arnberger and Haider, 2005; Arnberger and Eder, 2011); however, the preference for crowding differs slightly from the European context. Arnberger and Haider (2005), who also used an image-based DCE for a forest in Vienna, reported that both linear and quadratic parameters for the total number of people were statistically significant: low and high numbers exerted a negative impact on trail choices. In contrast, our results show that only the quadratic parameter is statistically significant: a low number of people is not a dominant factor, although their utility declined considerably at a high level of use. Our CLM results appear to be consistent with those of Arnberger et al. (2010) that the Japanese are generally tolerant of crowding, as are Asian people (Gillis et al., 1986; Homma, 1990). However, these discussions relate to the conditional logit model and reflect the preferences of the average respondent, whereas our LCM results indicate that segment 2 is much less tolerant of the number of people than segment 1. Revealing such detailed preferences, it is clear that LCM provides much richer information than CLM, from which only the “average” visitor’s preferences are obtained. We also discovered heterogeneous preferences for joggers and dogs, which confirms that urban forests fulfill various purposes and expectations of visitors. This information helps park managers to resolve problems of crowding and/or visitor conflicts.

4.2. Harvesting behavior on wild food plants

We found preference heterogeneity for watching plant pickers, whose presence was disliked by segment 1. This segment visited the park to enjoy the natural environment, including the

wild plants, and picking plants was therefore not appreciated. With regard to the perception of such harvesting behavior, it may be related to place attachment, which is a concept describing human–environmental interaction and containing emotional, cognitive, and functional bonds with specific places (Altman and Low, 1992; Jorgensen and Stedman, 2001). Previous research has found that visitors with higher place attachment are more sensitive to and responsible for social and environmental conditions (Kyle et al., 2004; Eder and Arnberger, 2012). Likewise, frequent visitors in segment 1 appear to have higher place attachment to the NFP and are therefore quite sensitive to harvesting behavior.

In contrast, the positive and significant parameter for plant picking in segment 2 was unexpected. The low frequency of visits by segment 2 suggests that respondents visit other recreational areas in addition to the NFP, possibly expecting those areas rather than the NFP to provide opportunities for appreciating a quality natural environment. One interpretation may be that these visitors accept consumptive recreational use in the NFP.

4.3. Implications for practical management

NFP visitors showed heterogeneous preferences for trail use. Information regarding which types of visitors use which areas of the park may help to reduce undesirable recreational experiences. Aikoh et al. (2010) found that visitors change their destinations and routes in the NFP depending on their trail preferences for nature walking. If visitors who dislike certain types of park activities (e.g., plant picking) have information about where those activities are more likely to occur, they can plan their visits accordingly to avoid such unpleasant encounters. This information could be particularly useful for first-time or infrequent visitors, who may not be aware of where they might encounter behaviors they would like to avoid.

The presence of dog-walkers, independently whether or not the dogs are leashed, negatively affected visitors' experiences in both segments. As mentioned above, walking with unleashed dogs is prohibited in the NFP. However, our results show that even encounters with leashed dogs detracts from visitors' satisfaction. Nevertheless, dog walking is an important activity, especially in residential areas, such as those that border the NFP. Thus, the introduction of visitor controls or daily, timely and/or seasonal restrictions may improve visitors' experiences. For example, the Osawa trail is famous for bird watching because the trail runs through old-growth forests where wild birds build their nests. In the NFP, their breeding season is limited from late spring to early summer. Thus, restricting dog walking on the Osawa trail during this period could effectively avoid conflicts between dog walkers and bird watchers.

One solution could be placing signs along the trail to inform and remind forest users that the park environment is to enjoy and not harvest; another solution may be putting up fences along those parts of the trail where people enter the woods to poach. Such actions could be especially effective in helping infrequent visitors to understand managers' warnings against plant picking, although some seasoned visitors will still pick plants regardless of these explicit appeals. Arnberger et al. (2010) has pointed out the difficulty in restricting certain types of activity in urban forests that are engrained in everyday life. If the above methods do not work, zonal restrictions on plant picking should be considered. Of course, an outright ban on plant picking in the NFP is an option, but this may result in a zero-sum conflict, which can only be resolved through sheer power politics (Eagles and McCool, 2004). Moreover, it may not be feasible for park managers to implement the ban across the NFP due to limited budgets and personnel.

4.4. Conclusion and limitations

Our contribution to the literature of urban forest management are two-fold. First, using the latent class approach, we identified heterogeneous preference on the use of an urban forest park along with the sources of the heterogeneity. Second, we shed light on plant pickers, who have not been examined much previously in the context of urban forests. We showed that preference toward plant pickers are heterogeneous and some users dislike seeing plant picking activities.

It is a global trend that more population is concentrated on urban areas. As it is difficult to have opportunities to experience solitude, it is possible that more and more people seek such an experience in an urban forests like those who fall in the segment 2 in our analysis. It is important for park managers to recognize heterogeneous preferences and manage parks accordingly. Our results also suggest that plant picking can potentially be an important source of conflicts in urban forests, which face ever-increasing demands for their services. To this end, we need to accumulate case studies on preferences towards plant picking activities. It is further warranted to examine the issue from historical and cultural perspectives to enhance our understanding of plant picking activities and thus avoid conflicts.

There are several limitations to our study. First, the image-based approach has some limits on how much information is conveyed. As we mentioned earlier, placing users overlapped on the two-dimensional space makes it hard for subjects to identify the attribute of each person in an image. This prohibited us from creating a situation with more people than the maximum we had for the experiment. Even though it is extremely rare to see so many people on a single trail in the NFP, it may have provided an interesting insight into how people would react to such a crowded situation, should it ever happen. The second limitation to any image-based experiments are potential discrepancies between what subjects perceive by looking at a picture and what they feel when they indeed are in the situation depicted by the picture. Choice experiments with 3D images and/or soundscape could address this limitation (Reichhart and Arnberger, 2010; Marin et al., 2011).

Another potential limitation that should be mentioned is our sampling. It should be noted that sampling is conducted at a section near the trailhead. While this study successfully elicit heterogeneous preferences for social conditions of trail use, it may be that our results are not representative of the other parts of NFP. The sample size was not very large even though statistical significance were observed for most of the variables. Finally, the majority of our samples are over 50 years old. While this is not due to a sampling bias as argued earlier, it warrants some caution in comparing our results with trail use preference in other countries where park visitors have considerably different age compositions.

Acknowledgement

The authors would like to acknowledge two anonymous referees for providing us with constructive comments and suggestions. Their comments brought substantial improvements to this study. We also would like to thank Koichi Kuriyama for his helpful advice regarding the choice design. A part of financial support for this study was provided by JSPS's Bilateral Joint Research Projects headed by Yoji Aoki.

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